Jimmy Dong

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EDUCATION

B.S. Computer Engineering Technology & Electrical Engineering Technology May '28

Rochester Institute of Technology | Rochester, NY

Coursework: Advanced Electronics, Signals & Systems, Hardware Description Language, Electronic Devices, Microcontroller Systems, Digital Systems Design, Circuits I & II.

SKILLS

Hardware: MSP430/MSP432, DE0-CV FPGA, soldering, prototyping, circuit analysis
Programming/HDL: C/C++, VHDL, Python, MATLAB, Java, R
Tools: Altium Designer, Quartus, ModelSim

PROJECTS

DEV1 Wire Harness Team Sep '25 - Present

RIT Electric Vehicle Team

- Integrated and routed wire harness systems to reduce signal interference and improve electrical reliability
- Performed continuity, voltage, and signal validation to verify harness integrity and electrical reliability.
- Assisted in diagnosing and resolving electrical faults, including connector failures and signal inconsistencies within drivetrain and control modules

Automated Texas Hold’em Card-Dispensing Robot (MSP432) Apr '26 - May '26

- Developed an MSP432-based card-dispensing system for Texas Hold’em using Embedded C and a state-machine structure to organize player, discard, flop, turn, and river operations.
- Integrated a TB6612FNG motor driver and DC feeder motor, configuring GPIO and TimerA PWM to control motor direction, speed, and card-feeding duration.
- Programmed and tested low-level GPIO, timer, PWM, and motor-control functions while troubleshooting hardware and firmware integration on the MSP432 platform.

Vending Machine Controller (Finite State Machine Design) Oct '25 - Dec '25

- Designed a finite state machine–based vending machine controller that accepts nickels, dimes, and quarters to dispense a $0.75 product with correct change handling
- Refined VHDL transaction-control logic to implement dispense validation, coin returns, and invalid-deposit handling.
- Displayed real-time balance using dual seven-segment displays, providing continuous user feedback throughout the transaction

Parallel Resonance Band-Pass Filter Design Sep '25 - Oct '25

- Designed and analyzed a parallel resonant band-pass filter using ideal and non-ideal inductor models to determine resonant frequency, Q factor, bandwidth, and voltage gain
- Performed AC frequency sweep simulation to identify resonance and 3 dB cutoff frequencies, comparing theoretical and simulated results
- Built and tested the physical circuit, collecting 50+ frequency response measurements to evaluate the impact of inductor series resistance on Q factor and bandwidth

7-Segment Display Controller (MSP430 Microcontroller) Feb '25 - Apr '25

- Programmed an MSP430 microcontroller in C to drive a multiplexed 4-digit seven-segment display via GPIO and timer-based refresh logic
- Executed digit extraction, timing control, and display refreshing algorithms for real-time numeric output without visible flicker
- Integrated switch inputs and validated display timing across multiple operating conditions